

Warm-start coordinate correction

Frozen plan v1 – execution 2026-09-12

Optimizer coordinates and null control. Let $B(\omega) = 2\sqrt{\omega(1-\omega)}X + (1-2\omega)Z$, so the matched Hamiltonian is $h_M(\omega) = -B(\omega)$. We use a common optimizer coordinate β with the cold gate $\exp(-i\beta X)$. The implemented warm gate is $\exp(-i\beta B(\omega))$; equivalently, its physical coordinate in $\exp(-i\beta_{\text{phys}}h_M)$ is $\beta_{\text{phys}} = -\beta$ at every objective evaluation. Its circuit matrix is $R_y(\theta_\omega)R_z(2\beta)R_y(-\theta_\omega)$, where $\theta_\omega = 2 \arcsin \sqrt{\omega}$. Setting every qubit probability, including auxiliaries, to 1/2 therefore recovers the cold path at the same parameters. The check covers depths one and two, native and auxiliary representations, and all policy paths. Setting only original probabilities to 1/2 does not define this control: the independence closure gives auxiliary probability 1/4.

Frozen correction study. We reran all 480 targeted tasks on the original 20 selected instances and completed all 4,320 original E4 tasks on its original 14 instances. Archived marginals, representations, seeds, and budgets were fixed. The execution comprises 4,332 fresh optimizations and 468 validated zero-layer initial-state records. Each fresh restart retains the original 60-evaluation COBYLA budget and early-stop accounting rule. Zero-layer records are not effective optimizations. We average three restarts, then five matched draws where applicable, before pairing policies within raw instances. Families, budgets, and studies are separate; intervals are unadjusted exploratory Student-t intervals with $n - 1$ degrees of freedom.

Targeted findings. On the fixed signal-enriched Max-3SAT subset ($n = 10$), pair-minus-cold differences are -6.7198 $[-9.2712, -4.1683]$ at $p = 1$ and -6.7971 $[-9.1395, -4.4547]$ at $p = 2$. Both spin-glass intervals include zero. Original-only and independence starts also improve over cold on this Max-3SAT subset. Pair-minus-original-only at $p = 2$ is -0.4759 $[-0.8454, -0.1063]$, while its $p = 1$ interval includes zero. Neither pair-minus-independence interval excludes zero.

Original E4 findings. The primary selected active-auxiliary subset retains two spin-glass and seven Max-3SAT instances. All eight pair-minus-cold intervals include zero. For Max-3SAT, the mean differences at $p = 1$, $p = 2$, $B = 128$, and $B = 256$ are -0.8717 , -0.8731 , -0.6768 , and -0.6218 , respectively. At $B = 128$, both selected spin-glass instances and two Max-3SAT instances have zero layers; they remain labelled in the all-scheduled summary. Positive-depth and zero-depth diagnostics are also reported separately. The Max-3SAT pair-minus-original-only contrast at $p = 2$ is -0.3108 $[-0.5562, -0.0654]$; this secondary exploratory contrast does not establish the primary pair-versus-cold claim.

Family	Budget	n	Mean difference	95% CI low	95% CI high
Spin glass	$p = 1$	10	0.0028	-0.2733	0.2789
Spin glass	$p = 2$	10	0.1153	-0.0468	0.2774
Max-3SAT	$p = 1$	10	-6.7198	-9.2712	-4.1683
Max-3SAT	$p = 2$	10	-6.7971	-9.1395	-4.4547

Table 1: Corrected pair-minus-cold original-objective differences on the fixed signal-enriched targeted set (10 raw instances per family). Means first average all three restarts. Negative differences favor pair warm starts. Intervals are unadjusted exploratory instance-level Student-t 95% intervals. All scheduled depths are retained, including zero-layer rows where applicable.

Family	Budget	n	Mean difference	95% CI low	95% CI high
Spin glass	$B = 128$	2	-0.2222	-3.0458	2.6014
Spin glass	$B = 256$	2	0.0374	-2.3424	2.4173
Spin glass	$p = 1$	2	-0.0037	-8.1718	8.1644
Spin glass	$p = 2$	2	-0.0915	-0.2520	0.0690
Max-3SAT	$B = 128$	7	-0.6768	-1.9673	0.6136
Max-3SAT	$B = 256$	7	-0.6218	-1.8792	0.6355
Max-3SAT	$p = 1$	7	-0.8717	-2.0089	0.2655
Max-3SAT	$p = 2$	7	-0.8731	-2.1540	0.4078

Table 2: Corrected pair-minus-cold original-objective differences on the original selected active-auxiliary subset (2 spin-glass and 7 Max-3SAT instances). Means first average all three restarts. Negative differences favor pair warm starts. Intervals are unadjusted exploratory instance-level Student-t 95% intervals. All scheduled depths are retained, including zero-layer rows where applicable.

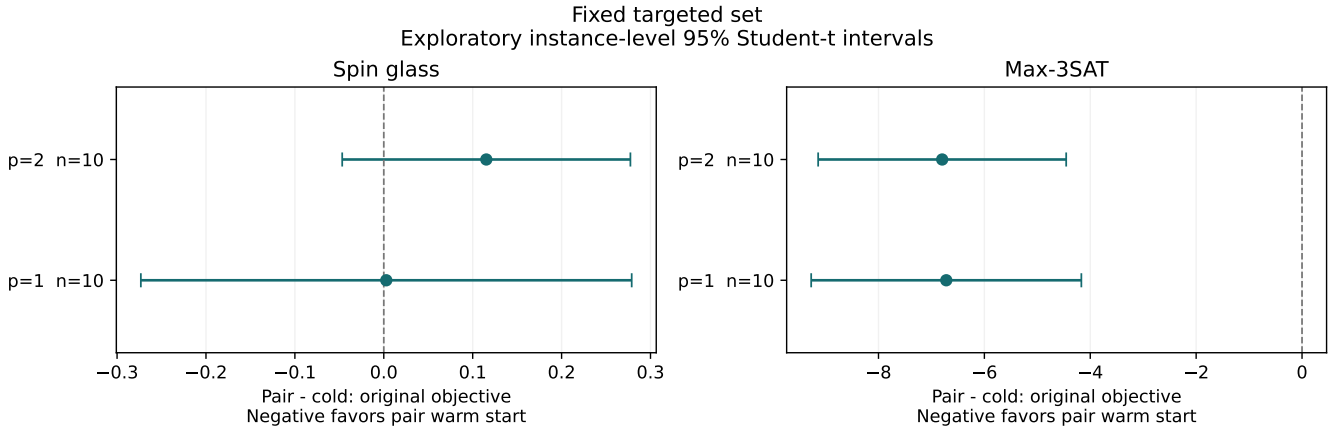


Figure 1: Fixed targeted study after coordinate correction; negative differences favor pair warm starts.

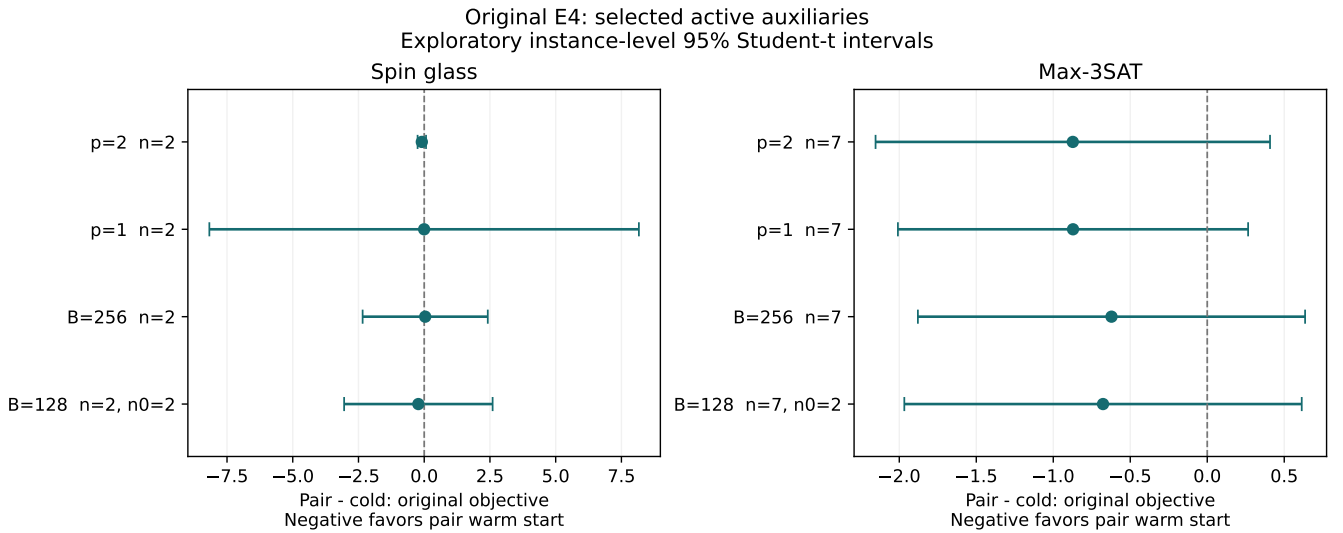


Figure 2: Original E4 selected active-auxiliary subset. The n_0 annotation counts zero-layer instances retained in the scheduled comparison.

Interpretation and provenance

The coordinate-corrected results support lower objectives relative to cold starts on the fixed signal-enriched Max-3SAT subset. They do not establish uniform improvement across families or the original E4 sample, nor a clear additional benefit of pair moments over the independence closure. Intervals containing zero indicate uncertain evidence, not equivalence or absence of any effect. The small original spin-glass active-auxiliary subset ($n = 2$) yields particularly wide intervals. All contrasts are exploratory and unadjusted for multiplicity.

With a common mixer optimizer coordinate, the targeted Max-3SAT improvement over cold starts remains, but the original E4 pair-versus-cold comparisons are inconclusive. These findings are limited to the frozen instances, representations, probabilities, and computational budgets; they do not imply a general advantage of auxiliary pair-informed starts.

Correction provenance. The frozen plan is `warmstart_coordinate_correction_plan_v1`; the execution is `exec_20260912_v2`. The local source snapshot is `1301a48c88d2`; its full hash, source manifest, environment snapshot, task attempts, and all 4,800 row identities are recorded in the accompanying audit. This snapshot is not an external GitHub release. The Linux/Python 3.12.14, NumPy 2.3.5, SciPy 1.17.0 environment differs from the Windows archive, so all positive-depth cold tasks were rerun uniformly. Changes from historical optimization results cannot be attributed solely to the mixer. Archived LP moments were restored and hash-checked without resolving the LP. Historical results were known when the plan was frozen; this is an exploratory correction study, not an outcome-blind confirmatory preregistration. Recomputing the archived original E4 rows with the same instance-level Student-t method also yields eight primary pair-minus-cold intervals containing zero. The change in claim strength must therefore not be attributed exclusively to the mixer correction.

Table 3: All five prespecified original-objective contrasts on the fixed selected active-auxiliary subsets. A: targeted; B: original E4; SG: spin glass; SAT: Max-3SAT. Intervals are unadjusted instance-level Student-t 95% intervals; n_0 counts instances with a zero-layer draw.

Study/family	Budget	Contrast	n	Mean [95% CI]	n_0
B/SG	B=128	Independence - cold	2	-0.2222 [-3.0458, 2.6014]	2
B/SG	B=128	Original-only - cold	2	-0.2222 [-3.0458, 2.6014]	2
B/SG	B=128	Pair - cold	2	-0.2222 [-3.0458, 2.6014]	2
B/SG	B=128	Pair - independence	2	0.0000 [-0.0000, 0.0000]	2
B/SG	B=128	Pair - original-only	2	0.0000 [0.0000, 0.0000]	2
B/SG	B=256	Independence - cold	2	-0.0748 [-4.9278, 4.7783]	0
B/SG	B=256	Original-only - cold	2	0.0699 [-2.7225, 2.8623]	0
B/SG	B=256	Pair - cold	2	0.0374 [-2.3424, 2.4173]	0
B/SG	B=256	Pair - independence	2	0.1122 [-2.3610, 2.5854]	0
B/SG	B=256	Pair - original-only	2	-0.0325 [-0.4450, 0.3801]	0
B/SG	p=1	Independence - cold	2	-0.3163 [-4.9533, 4.3206]	0
B/SG	p=1	Original-only - cold	2	-0.0378 [-8.6400, 8.5643]	0
B/SG	p=1	Pair - cold	2	-0.0037 [-8.1718, 8.1644]	0
B/SG	p=1	Pair - independence	2	0.3127 [-3.2185, 3.8439]	0
B/SG	p=1	Pair - original-only	2	0.0342 [-0.3999, 0.4682]	0
B/SG	p=2	Independence - cold	2	-0.0523 [-2.8029, 2.6984]	0
B/SG	p=2	Original-only - cold	2	0.3283 [-4.8454, 5.5020]	0
B/SG	p=2	Pair - cold	2	-0.0915 [-0.2520, 0.0690]	0
B/SG	p=2	Pair - independence	2	-0.0393 [-2.9504, 2.8719]	0
B/SG	p=2	Pair - original-only	2	-0.4198 [-5.7540, 4.9144]	0
B/SAT	B=128	Independence - cold	7	-0.8082 [-2.0830, 0.4665]	2
B/SAT	B=128	Original-only - cold	7	-0.6470 [-1.9346, 0.6406]	2
B/SAT	B=128	Pair - cold	7	-0.6768 [-1.9673, 0.6136]	2
B/SAT	B=128	Pair - independence	7	0.1314 [-0.3257, 0.5885]	2
B/SAT	B=128	Pair - original-only	7	-0.0298 [-0.1019, 0.0422]	2
B/SAT	B=256	Independence - cold	7	-0.4567 [-2.0344, 1.1209]	0
B/SAT	B=256	Original-only - cold	7	-0.5817 [-1.7950, 0.6316]	0
B/SAT	B=256	Pair - cold	7	-0.6218 [-1.8792, 0.6355]	0
B/SAT	B=256	Pair - independence	7	-0.1651 [-0.8271, 0.4968]	0
B/SAT	B=256	Pair - original-only	7	-0.0401 [-0.2648, 0.1845]	0
B/SAT	p=1	Independence - cold	7	-0.5523 [-1.8724, 0.7679]	0

Study/family	Budget	Contrast	n	Mean [95% CI]	n_0
B/SAT	p=1	Original-only - cold	7	-0.5616 [-1.7635, 0.6402]	0
B/SAT	p=1	Pair - cold	7	-0.8717 [-2.0089, 0.2655]	0
B/SAT	p=1	Pair - independence	7	-0.3195 [-0.8106, 0.1717]	0
B/SAT	p=1	Pair - original-only	7	-0.3101 [-0.6714, 0.0511]	0
B/SAT	p=2	Independence - cold	7	-0.8244 [-2.0359, 0.3871]	0
B/SAT	p=2	Original-only - cold	7	-0.5623 [-1.7411, 0.6164]	0
B/SAT	p=2	Pair - cold	7	-0.8731 [-2.1540, 0.4078]	0
B/SAT	p=2	Pair - independence	7	-0.0487 [-0.4092, 0.3117]	0
B/SAT	p=2	Pair - original-only	7	-0.3108 [-0.5562, -0.0654]	0
A/SG	p=1	Independence - cold	10	-0.0171 [-0.3315, 0.2973]	0
A/SG	p=1	Original-only - cold	10	0.1118 [-0.1106, 0.3343]	0
A/SG	p=1	Pair - cold	10	0.0028 [-0.2733, 0.2789]	0
A/SG	p=1	Pair - independence	10	0.0199 [-0.2391, 0.2790]	0
A/SG	p=1	Pair - original-only	10	-0.1090 [-0.3286, 0.1106]	0
A/SG	p=2	Independence - cold	10	0.1168 [-0.3115, 0.5450]	0
A/SG	p=2	Original-only - cold	10	0.1351 [-0.1799, 0.4501]	0
A/SG	p=2	Pair - cold	10	0.1153 [-0.0468, 0.2774]	0
A/SG	p=2	Pair - independence	10	-0.0015 [-0.3104, 0.3075]	0
A/SG	p=2	Pair - original-only	10	-0.0198 [-0.2369, 0.1974]	0
A/SAT	p=1	Independence - cold	10	-6.8015 [-9.3293, -4.2737]	0
A/SAT	p=1	Original-only - cold	10	-6.6590 [-9.0771, -4.2409]	0
A/SAT	p=1	Pair - cold	10	-6.7198 [-9.2712, -4.1683]	0
A/SAT	p=1	Pair - independence	10	0.0818 [-0.0328, 0.1963]	0
A/SAT	p=1	Pair - original-only	10	-0.0608 [-0.2766, 0.1551]	0
A/SAT	p=2	Independence - cold	10	-6.6320 [-9.0352, -4.2288]	0
A/SAT	p=2	Original-only - cold	10	-6.3212 [-8.6583, -3.9841]	0
A/SAT	p=2	Pair - cold	10	-6.7971 [-9.1395, -4.4547]	0
A/SAT	p=2	Pair - independence	10	-0.1651 [-0.6731, 0.3429]	0
A/SAT	p=2	Pair - original-only	10	-0.4759 [-0.8454, -0.1063]	0